

28 - Functional Data Analysis: Scope and Examples.

Hilbert Spaces for Functional Data Embedding.

Smoothing Functional Data

Applied Statistics

Unat Teksen

May 18, 2026

1 What is Functional Data Analysis?

1.1 Core Idea

- In classical statistics, a data point is a **number or a vector** (e.g. a person's height is one number).
- In **Functional Data Analysis (FDA)**, a data point is an **entire function (curve, surface, or image)**.
- Instead of studying “what is the temperature today?”, FDA studies “what does the entire temperature curve over one year look like for each city?” — and treats that whole curve as *one* data point.
- **Also called:** object-oriented data analysis.

1.2 Motivating Example

- Suppose we measure the height of $n = 10$ girls at $p = 31$ different ages.
- Here $p = 31 \gg n = 10$: classical statistics breaks down.
- But the 31 measurements are *not* 31 separate variables — they come from **one continuous underlying growth process**.
- Treating them as a function lets us:
 - Compute **derivatives** (how fast is the child growing? Is growth accelerating?).
 - **Align** curves (one child hits puberty at age 11, another at 13 — time-warp to compare them).
 - Apply multivariate methods in the function space.

1.3 Typical Goals of FDA

- **Represent** data in ways that highlight salient features (display, summarise).
- **Study main sources of pattern and variation** among the curves.
- **Explain an outcome** (response) using the function as a predictor.
- **Classify or compare groups** of curves with respect to certain types of variation.

1.4 Course Roadmap

1. Hilbert space model for functional data.
2. Smoothing and interpolation of functional data.
3. FDA and dimensionality reduction in Hilbert spaces (functional PCA).
4. Data alignment and clustering.

2 Hilbert Space Model for Functional Data

2.1 Why Do We Need a Space for Functions?

- For regular vectors in $\mathbb{R}^p$ we know how to compute distances, averages, and projections — we have **geometry**.
- To do the same for functions, we embed them in a space that also has geometry: a **Hilbert space**.
- In a Hilbert space, the points *are* functions, so many multivariate methods extend naturally.

2.2 Definition: Hilbert Space

- A **real Hilbert space** H is an inner-product space that is **complete** in the norm induced by the inner product (no “gaps” — every Cauchy sequence converges inside H).
- H is **separable** if it contains a dense countable subset.
- If H is separable $\Rightarrow$ it has an **ortho-normal basis** $\{u_n\}$, and any element can be written:

$$x = \sum_{n=1}^{\infty} \langle x, u_n \rangle u_n$$

- The inner product gives us: **length** $\|f\| = \sqrt{\langle f, f \rangle}$, **angle**, **orthogonality** ($\langle f, g \rangle = 0$), and **projections**.

2.3 The L^2 Space

- The most important example: the space of **square-integrable functions** on interval I :

$$L^2(I) = \left\{ f : I \rightarrow \mathbb{R} \mid \int_I |f(t)|^2 dt < \infty \right\}$$

- Inner product** in L^2 :

$$(f_1, f_2) = \int_I f_1(t) \cdot f_2(t) dt$$

This is the natural generalisation of the dot product $\sum_i a_i b_i$ — sum becomes integral.

- L^2 is technically a quotient space: two functions are *identical* if they differ only on a set of measure zero:

$$x = y \quad \text{if} \quad \int_I |x(t) - y(t)|^2 dt = 0$$

2.4 The Bayes Hilbert Space B^2 (for Density Functions)

- L^2 geometry **does not work** for probability density functions: if you add two densities in L^2 , the result does not integrate to 1 — the geometry is wrong.
- B^2 is the space of density functions on a closed interval I , with inner product:

$$\langle f, g \rangle_B = \frac{1}{2\eta} \int_I \int_I \ln\left(\frac{f(t)}{f(s)}\right) \ln\left(\frac{g(t)}{g(s)}\right) dt ds$$

where $\eta = |I|$ is the length of the interval.

- Perturbation** (“addition” of two densities):

$$(f \oplus g)(t) = \frac{f(t) g(t)}{\int_I f(s) g(s) ds}$$

This is exactly the **Bayesian update rule** — combining prior f and likelihood g gives the posterior.

- Powering** (scalar multiplication by α):

$$(\alpha \odot f)(t) = \frac{f(t)^\alpha}{\int_I f(s)^\alpha ds}$$

- Note:** Exponential families of distributions are simply *linear spaces* in B^2 .

2.5 The Centred Log-Ratio (clr) Transformation

- B^2 is **isomorphic** to L^2 , so we can always convert density data into L^2 using the **clr** transformation:

$$\text{clr}(f)(t) = f_c(t) = \ln(f(t)) - \frac{1}{\eta} \int_I \ln(f(s)) ds$$

- Subtract the mean of $\ln f$ from itself $\Rightarrow$ zero-mean function in L^2 .
- After applying clr , all standard L^2 tools (PCA, regression, etc.) can be used on density data.

2.6 Choosing the Embedding Space

- The embedding space must match **the properties of the data** and **the goal of the analysis**.
- Density functions $\rightarrow$ use B^2 .
- Care about derivatives/smoothness $\rightarrow$ use **Sobolev spaces** (derivatives enter the norm).
- This is a **modelling decision**, not automatic.

2.7 Functional Random Variables and Data Sets

Let H be a Hilbert space whose points are functions on $T = [t_{\min}, t_{\max}]$.

- **Functional random variable**: a random element $X : \Omega \rightarrow H$ — for each outcome ω you get a function $X(\omega) : T \rightarrow \mathbb{R}$.
- **Functional datum**: one realisation $x = X(\omega) : T \rightarrow \mathbb{R}$ — one observed curve.
- **Functional data set**: a collection of N realisations $x_1, x_2, \dots, x_N$.

2.8 The Fréchet Mean

- The mean of X in H (assuming $\mathbb{E}[\|X\|_H^4] < \infty$) is the unique $\mu \in H$ solving:

$$\mu = \arg \min_{x \in H} \mathbb{E}[\|X - x\|_H^2]$$

- Special cases:
 - If $H = L^2$: $\mu(t) = E[X(t)]$ — pointwise mean function.
 - If $H = B^2$: $\mu = \text{clr}^{-1}(\mathbb{E}[\text{clr}(X)])$.
- **Note**: $E[X](t) = E[X(t)]$ so for L^2 we can forget the Fréchet definition and just take pointwise expectations.
- **Sample estimator** (in any H):

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$$

The meaning of “sum” depends on H — in B^2 it is a sequence of perturbations $\oplus$.

2.9 Co-variance Operator

Let $X : \Omega \rightarrow H$ be a **zero-mean** functional random variable with $\mathbb{E}[\|X\|_H^4] < \infty$.

- **Definition — co-variance operator:** the operator $C_x : H \rightarrow H$ defined as:

$$C_x = \mathbb{E}[\langle X, x \rangle X], \quad x \in H$$

- Must be **symmetric**, **positive semi-definite**, and the sum of eigenvalues must be finite $\Rightarrow C_x$ is a **Hilbert-Schmidt operator**.
- If $H = \mathbb{R}^p$: C_x coincides with the covariance matrix $\Sigma = \mathbb{E}[XX^T]$.
- If $H = L^2$: C_x is expressed via a **kernel**:

$$[Cx](t) = \int_I c(s, t) x(s) ds, \quad x \in L^2$$

where the **co-variance kernel** is the pointwise covariance:

$$c(s, t) = \mathbb{E}[X(s) X(t)]$$

- **Sample co-variance operator** (in any H):

$$Sx = \frac{1}{N} \sum_{i=1}^N \langle X_i, x \rangle X_i, \quad x \in H$$

- In L^2 , with sample kernel $\hat{c}(s, t) = \frac{1}{N} \sum_{i=1}^N X_i(s) X_i(t)$:

$$[Sx](t) = \int_T \hat{c}(s, t) x(s) ds$$

- **Note:** In the heat-map of the sample covariance kernel, the main diagonal runs from bottom-left to top-right (nearby time points are highly correlated).

3 Smoothing and Interpolation of Functional Data

3.1 The Setup

- In practice, you never observe a complete continuous function — you observe **discrete, noisy measurements**.
- For function x_i , you observe n_i pairs (t_{ij}, y_{ij}) with $j = 1, \dots, n_i$.
- Model:

$$y_j = x(t_j) + \epsilon_j$$

where x is the **unknown true function** and ϵ_j is measurement noise.

- **Goal:** recover x from the pairs (t_j, y_j) — a regression problem where the estimand is a function.

3.2 Interpolation vs. Smoothing

- **Interpolation:** the reconstructed function passes *exactly* through every observed point.
 - Appropriate when measurements are **noiseless**.
 - With noise: fits the noise too, and **derivatives explode**.
- **Smoothing:** the reconstructed function is smoother than the raw observations.
 - Appropriate when measurements are **noisy**.
 - Derivatives remain stable and meaningful.
- **Smoothing is usually preferred** because interpolation amplifies noise in derivatives.

3.3 Basis Function Representation

- Choose K known **basis functions** $\phi_1, \dots, \phi_K$ and represent x as their weighted sum:

$$x(t) = \sum_{k=1}^K c_k \phi_k(t) = \underline{\phi}^T \underline{c}$$

where $\underline{\phi} = [\phi_1, \dots, \phi_K]^T$ and $\underline{c} = [c_1, \dots, c_K]^T$ are the **coefficients to estimate**.

- In a separable Hilbert space with ortho-normal basis $\{u_n\}$:

$$x = \sum_{n=1}^K \langle x, u_n \rangle u_n$$

- The basis functions are like musical notes — any melody (function) is a combination of basic sounds ϕ_k with weights c_k .
- **Key question: how large should K be?**
 - Too small $\Rightarrow$ miss features (underfitting, high bias).
 - Too large $\Rightarrow$ overfit noise (high variance, derivative instability).

3.4 Choice of Basis Functions

3.4.1 Fourier Basis

- Good for **periodic phenomena**.
- Robust: small variance, but can be biased.
- Fast to compute (Fast Fourier Transform).
- **Cannot** handle discontinuous functions.

$$\hat{x}(t) = c_0 + c_1 \sin(\omega t) + c_2 \cos(\omega t) + \dots$$

$$\phi_0(t) = 1, \quad \phi_{2r-1}(t) = \sin(r\omega t), \quad \phi_{2r}(t) = \cos(r\omega t)$$

3.4.2 Splines

- Used for **non-periodic, possibly discontinuous data**.
- To construct an m -order spline on T :
 1. Divide T into L sub-intervals by fixing a set of **knots**.
 2. On each sub-interval, define a polynomial of order m (m constants needed).
 3. Polynomials connect with continuity up to order $m - 2$ of their derivatives at knots.
- Total parameters (non-overlapping knots): $m + L - 1$.
- To fit a **discontinuity**: place two knots on top of each other.
- **B-splines** (basis splines) are most common — each basis function is **localised** (non-zero only in a small region of T).
- To increase flexibility: add more knots where the function varies more.

3.5 OLS Smoothing

- Model: $y_j = \sum_{k=1}^K c_k \phi_k(t_j) + \epsilon_j$.
- **Design matrix** ($n \times K$):

$$\Phi = \begin{bmatrix} \phi_1(t_1) & \cdots & \phi_K(t_1) \\ \vdots & & \vdots \\ \phi_1(t_n) & \cdots & \phi_K(t_n) \end{bmatrix}$$

- **Objective** — minimise sum of squared errors:

$$SMSSE(\underline{y} | \underline{c}) = \sum_{j=1}^n \left[y_j - \sum_{k=1}^K c_k \phi_k(t_j) \right]^2$$

- **OLS solution:**

$$\begin{aligned} \hat{\underline{c}} &= (\Phi^T \Phi)^{-1} \Phi^T \underline{y} \\ \hat{\underline{y}} &= \Phi \hat{\underline{c}} \end{aligned}$$

- $\hat{\underline{y}}$ is the **orthogonal projection** of $\underline{y}$ onto the column space of Φ — identical to ordinary linear regression with $\underline{c}$ playing the role of β .
- OLS is appropriate when errors are **independent and identically distributed (i.i.d.)**.

3.6 Variance of the Estimates

- **Variance of coefficient estimator** (general):

$$\text{Var}[\underline{c}] = (\Phi^T W \Phi)^{-1} \Phi^T W \Sigma_e \Phi (\Phi^T W \Phi)^{-1}$$

- For OLS with i.i.d. errors:

$$\text{Var}[\underline{c}] = \sigma^2 (\Phi^T \Phi)^{-1}$$

- **Variance of pointwise curve estimate:**

$$\text{Var}[\hat{y}] = \Phi \text{Var}[\underline{c}] \Phi^T$$

For OLS:

$$\text{Var}[\hat{y}] = \sigma^2 \Phi (\Phi^T \Phi)^{-1} \Phi^T = \sigma^2 S$$

where $S = \Phi (\Phi^T \Phi)^{-1} \Phi^T$ is the **hat (projection) matrix**.

- **Residual variance estimate:**

$$s^2 = \frac{1}{n - K} \sum_j (y_j - \hat{y}_j)^2$$

(n observations minus K estimated parameters.)

3.7 GLS Smoothing

- If errors are **not i.i.d.** (correlated or heteroscedastic), use **Generalised Least Squares (GLS)**.
- Minimise the **weighted** sum of squared errors:

$$SMSSSE(\underline{y} | \underline{c}) = (\underline{y} - \Phi \underline{c})^T W (\underline{y} - \Phi \underline{c})$$

where W is a positive definite weight matrix (e.g. inverse of the error covariance).

- **GLS solution:**

$$\hat{\underline{c}} = (\Phi^T W \Phi)^{-1} \Phi^T W \underline{y}$$

3.8 Choosing K — Bias-Variance Trade-off

- **Large K** : small bias (fits data well) but **high variance** (overfits noise, derivatives explode).
- **Small K** : large bias (misses features) but **small variance** (stable).
- Strategy: look for the value of K where **sampling variance drops sharply** (a “knee” in the variance curve).
- Can use step-wise variable selection or cross-validation.

3.9 Penalised Smoothing

- Instead of choosing K discretely, add a **roughness penalty** that punishes highly curved functions.
- **Second-derivative roughness penalty** (measuring curvature):

$$\text{Pen}_2(x) = \int [D^2 x(s)]^2 ds$$

Zero for a straight line; large for a wiggly function.

- **Penalised SSE** with smoothing parameter λ :

$$PenSSE_{\lambda}(x | \underline{y}) = [\underline{y} - x(\underline{t})]^T W [\underline{y} - x(\underline{t})] + \lambda Pen_2(x)$$

- Role of λ :
 - $\lambda \rightarrow \infty$: heavy penalisation $\Rightarrow$ fitted curve is a **straight line**.
 - $\lambda \rightarrow 0$: no penalisation $\Rightarrow$ curve **interpolates** the data.
- **Note:** The minimiser of $PenSSE$ with D^2 penalty is a **cubic spline** with knots at the data points t_j .
- In general, for an m -th order derivative penalty:

$$Pen_m(x) = \int [D^m x(s)]^2 ds = \underline{c}^T R \underline{c}$$

where the **roughness matrix** is:

$$R = \int D^m \phi(s) D^m \phi^T(s) ds$$

- Full penalised objective in matrix form:

$$PenSSE_m(\underline{y}, \underline{c}) = [\underline{y} - \Phi \underline{c}]^T W [\underline{y} - \Phi \underline{c}] + \lambda \underline{c}^T R \underline{c}$$

- **Penalised solution:**

$$\hat{\underline{c}} = (\Phi^T W \Phi + \lambda R)^{-1} \Phi^T W \underline{y}$$

- **Note:** When $R = I$, this is exactly **ridge regression**.
- **Remark:** Embedded space and smoothing are much related — the penalty chosen implicitly defines the geometry of the function space (e.g. D^2 penalty corresponds to a Sobolev space).

3.10 Choosing λ — Generalised Cross-Validation (GCV)

- The fitted values can be written as:

$$\hat{\underline{y}} = \Phi(\Phi^T W \Phi + \lambda R)^{-1} \Phi^T W \underline{y} = S_{\phi, \lambda} \underline{y}$$

where $S_{\phi, \lambda}$ is the **smoothing (hat) matrix**.

- $\text{trace}(S_{\phi, \lambda})$ = effective number of parameters (model complexity).
- **GCV criterion:**

$$GCV(\lambda) = n^{-1} SSE \cdot \frac{1}{[n^{-1} \text{trace}(I - S_{\phi, \lambda})]^2}$$

- Choose λ that **minimises** $GCV(\lambda)$ — balances fit quality (SSE) against complexity (trace).

Türkçe Özet — Basit Anlatım

Genel Soru: Ne Yapmaya Çalışıyoruz?

- Elimizde gerçek hayattan gelen veriler var, ama bunlar normal “sayı listeleri” değil — **eğriler, zaman boyunca değişen süreçler**.
- Örnek: 10 kız çocuğunun 31 yıl boyunca ölçülen boy uzunlukları.
- Tüm teknikler şu soruya cevap arıyor: *“Bu eğri verisini nasıl temsil ederim, nasıl analiz ederim, nasıl gürültüden temizlerim?”*

1. Fonksiyonel Veri Nedir?

- Normal istatistikte veri noktası = bir sayı ya da birkaç sayı.
- FDA’da veri noktası = **bir fonksiyon (bir eğri)**.
- 10 kişimiz varsa, 10 adet eğrimiz var — her eğri birer “veri noktası.”
- Neden bunu yapıyoruz?
 - Eğrilerin **türevini alabiliriz** (büyüme hızı ne zaman en yüksek?).
 - Eğrileri **hizalayabiliriz** (bir çocuk ergenliğe 11 yaşında, diğeri 13 yaşında girdi — karşılaştırmak için hizalamamız lazım).
 - Eğrilerle normal istatistik yöntemlerini uygulayabiliriz.

2. Hilbert Uzayı: “Eğrilerin Yaşadığı Yer”

- Normal sayılar $\mathbb{R}^3$ gibi bir uzayda yaşar. Eğriler için özel bir uzay lazım: **Hilbert Uzayı**.
- Okulda öğrendiğin iç çarpım (dot product): $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$.
- Eğriler için aynı şey ama toplam yerine **integral**:

$$\langle f, g \rangle = \int f(t) \cdot g(t) dt$$

- Bu iç çarpım sayesinde: eğrinin uzunluğunu, iki eğrinin dikliğini, ortalama eğriyi hesaplayabilirsin.
- **L^2 uzayı**: Kare integrallenebilir fonksiyonlar. Sıradan eğriler için kullanılır.
- **B^2 uzayı**: Olasılık yoğunluk fonksiyonları için. L^2 burada işe yaramaz çünkü iki yoğunluk fonksiyonunu L^2 ’de toplarsan 1’e integral vermeyen bir şey çıkar.
- **Uzay seçimi bir modelleme kararıdır** — verinin türüne ve analizin amacına göre değişir.

B^2 için clr Dönüşümü

- $B^2 \cong L^2$ olduğundan, yoğunluk verilerini **centred log-ratio (clr)** ile L^2 'ye dönüştürebiliriz:

$$\text{clr}(f)(t) = \ln(f(t)) - \frac{1}{\eta} \int_I \ln(f(s)) ds$$

- Fonksiyonun logaritmasını al, ortalamasını çıkar $\Rightarrow L^2$ 'de sıfır-ortalıklı fonksiyon.
- Bu dönüşümden sonra normal L^2 araçlarını kullanabilirsin.

3. Ortalama ve Kovaryans — Eğrilerin İstatistiği

- **Ortalama eğri (Fréchet Ortalaması):** Tüm eğrilere en yakın olan eğriyi bul:

$$\mu = \arg \min_{x \in H} \mathbb{E}[\|X - x\|_H^2]$$

L^2 'de bu sadece şu demek: her zaman noktasında eğrilerin ortalamasını al, $\mu(t) = E[X(t)]$.

- **Örnek:** 10 kişinin 5. yaşındaki boylarının ortalaması, 6. yaşındakilerin ortalaması... bunları birleştiren ortalamaya büyüme eğrisini elde edersin.
- **Kovaryans Operatörü** (L^2 'de):

$$[Cx](t) = \int_I c(s, t) x(s) ds$$

Burada $c(s, t) = \mathbb{E}[X(s) \cdot X(t)]$ — s ve t zamanlarındaki değerlerin birlikte nasıl değiştiği.

- **Örnek:** Çocuğun 5 yaşındaki boyu uzunsa, 10 yaşında da uzun mu? $c(5, 10)$ bunu söyler.

4. Smoothing — Gürültüyü Temizle

- Gerçekte elimizde temiz eğriler yok — sadece **gürültülü ölçümler** var:

$$y_j = x(t_j) + \epsilon_j$$

- **İnterpolasyon:** Eğri her noktadan tam geçer. Gürültülü veride türev alınca çığırılır.
- **Smoothing:** Eğri genel trendi yakalar. Türev almak güvenli. *Genellikle tercih edilir.*
- **Baz fonksiyonları** ile eğriyi ifade et:

$$x(t) = \sum_{k=1}^K c_k \phi_k(t)$$

Tıpkı şuna benziyor: herhangi bir müziği “Do, Re, Mi” gibi temel seslerin kombinasyonu olarak ifade et.

- **OLS ile katsayıları bul:**

$$\hat{\underline{c}} = (\Phi^T \Phi)^{-1} \Phi^T \underline{y}, \quad \hat{\underline{y}} = \Phi \hat{\underline{c}}$$

- **K seçimindeki denge:**

- Büyük $K \Rightarrow$ az bias, çok varyans (overfit).
- Küçük $K \Rightarrow$ çok bias, az varyans (underfit).

5. Penalized Smoothing — Pürüzlülüğü Cezalandır

- K seçmek yerine: eğrinin **pürüzlülüğünü cezalandır**.
- İkinci türev pürüzlülüğü ölçer (eğrinin “kıvrımı”):

$$Pen_2(x) = \int [D^2 x(s)]^2 ds$$

Düz çizgi için sıfır; kıvrımlı eğri için büyük.

- Minimize edilen:

$$PenSSE_\lambda = (\text{veri uyum terimi}) + \lambda \cdot Pen_2(x)$$

- **λ rolü:**

- $\lambda \rightarrow \infty \Rightarrow$ eğri düz çizgiye dönüşür.
- $\lambda \rightarrow 0 \Rightarrow$ eğri her noktadan geçer (interpolasyon).

- **Çözüm:**

$$\hat{\underline{c}} = (\Phi^T W \Phi + \lambda R)^{-1} \Phi^T W \underline{y}$$

$R = I$ olduğunda bu tamamen **ridge regresyon** olur.

- **λ 'yı otomatik seç — GCV:**

$$GCV(\lambda) = n^{-1} SSE \cdot \frac{1}{[n^{-1} \text{trace}(I - S_{\phi, \lambda})]^2}$$

En küçük $GCV(\lambda)$ 'yı veren λ 'yı seç.

Sınav İçin En Kritik Sezgi Noktaları

- **Neden Hilbert uzayı?** Eğrileri “nokta” gibi davranabilmek için. Uzayı seçmek = veriyle ne yapmak istediğini tanımlamak.
- **L^2 vs B^2 :** Yoğunluk fonksiyonları için L^2 kullanamayız çünkü iki yoğunluğu L^2 'de toplarsan ortaya çıkan şey yoğunluk fonksiyonu olmaz (1'e integral vermez).
- **Smoothing vs interpolasyon:** Gürültülü veride interpolasyon yaparsan türev alınca eğri deliriyor. Smoothing daha güvenli.
- **K büyük/küçük dengesi:** Büyük $K \rightarrow$ az bias, çok varyans. λ (penalization) da aynı şeyi farklı bir dille söylüyor: λ büyüdükçe eğri düzleşir.
- **Embedded space ve smoothing bağlantısı:** Hangi penalty kullandığın, hangi Hilbert uzayında çalıştığını belirler. Bu iki karar ayrı değil — aynı madalyonun iki yüzü.